Structured Becoming

A Revisable Relational Account of Persistence, Change, and Dependent Causation

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4 August 2026
WORKING PAPER - VERSION 0.3

This paper proposes a candidate framework for understanding how natural systems persist while changing. It does not claim that reality literally is a graph, computation, circuit, manifold, state vector, or structured occurrence. The categories used here are revisable tools for modeling natural persistence, not declarations of an absolute ontology.

Persistence without permanence. Organization without isolation. Precision without reification.

Abstract

What makes a thing continue as that thing while its components, properties, and relations change? Structured Becoming answers with a relational account of persistence. At a selected scale, a natural system is represented by its currently realized state, its relational organization, and the dependencies through which both are sustained. Persistence is not exact sameness and does not require an immutable bearer. It is causally traceable, organizationally continuous carryover through transformation. The proposal is realist about natural systems but modest about models: the system is not identical with its description, and no formal decomposition is assumed to be unique, final, or privileged. State, structure, boundary, whole, relation, and emptiness are all treated as revisable distinctions rather than self-existing foundations. The paper distinguishes constitution from description, separates natural systems from observations and models, and leaves observer dependence and Bell-type nonlocality explicitly unresolved. A compact formal appendix retains the original framework's state-structure equations, persistence criteria, RAIL, SHADOW, KEEPER logic, and structure-state alignment result, while the main text develops the philosophical argument in prose.

Keywords: persistence; process; relational ontology; state and structure; dependent causation; identity; apophatic restraint; constitution and description.

Terminological note

This paper avoids using "intrinsic nature" for a system's current internal organization. In ordinary English, intrinsic can mean constitutive or internal. In Madhyamaka, however, intrinsic nature is often used technically for an independent, self-grounding nature. To prevent ambiguity, the present paper uses constitutive organization for the former and reserves intrinsic self-nature for the latter.

1. The problem: persistence without permanence

Natural systems change continuously, yet many of them remain identifiable across time. Organisms metabolize, storms exchange matter and energy, institutions replace members, and computations update state. Exact sameness would make change incompatible with persistence. Mere reuse of a name would make identity arbitrary. A satisfactory account must explain continuity without requiring an unchanging substance and without reducing persistence to language alone.

The central proposal is that a thing persists through a causally and relationally continuous organization of state, structure, and dependencies. These terms do not name independent ingredients placed side by side. They distinguish aspects of one ongoing occurrence. The state is meaningful only within an organization of possible distinctions and transitions; the structure is realized only through actual states and interactions; dependencies connect both to prior history, neighboring systems, and larger constraints.

At any moment, the current state-structure-dependency configuration is the leading edge of the system's existence. It is what is presently available to be transformed into a successor. This does not imply that an essence travels unchanged through time. What continues is an organized causal trajectory whose criteria of continuity may differ by kind, scale, and research purpose.

Central thesis

A thing persists when its present organization remains causally available to a successor and enough of the organization relevant to its identity is carried forward under the active local and global conditions. Persistence is organized continuity through change, not permanence beneath change.

2. Scope, realism, and the status of the model

Structured Becoming makes a modest realist commitment: natural systems and their causal organization do not depend on any particular observer, description, or formal model in order to exist. A cell does not begin to metabolize because it is represented in a diagram. A bridge does not acquire load-bearing relations because an engineer writes equations. Natural organization constrains what can occur and therefore constrains which descriptions can succeed.

The framework itself is not objective reality. It is an abstract representation constructed from a perspective, at a selected scale, with declared boundaries, variables, and tolerances. It can be tested, corrected, compared with rival models, and made increasingly accurate. It cannot become the natural system it represents. The distinction is not a defect; it is the condition of disciplined inquiry.

The paper therefore separates four items that are often conflated: the natural system; an observation or measurement produced through an interaction; a model constructed from those records; and a metaphysical interpretation of the model. Evidence may constrain each successive level, but no level is simply identical with the next.

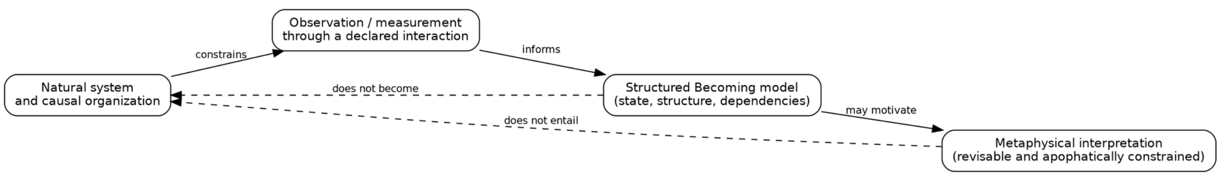

Figure 1. Natural system, observation, model, and interpretation are connected but not identical. Formal success does not turn a model into objective reality or entail a final ontology.

This separation also prevents two opposite errors. The first is naive reification: treating a useful formal object as a self-existing constituent of reality. The second is indiscriminate subjectivism: treating every description as equally arbitrary because all models are constructed. Natural systems push back. Some models predict, intervene, and generalize better than others because they track real regularities, even though no model exhausts the system.

3. The structured occurrence

The candidate unit of analysis is a structured occurrence: a presently realized stage in an inherited trajectory. It is not an inert bearer that first exists and then acquires relations. Its existence is already relationally organized. For exposition, the occurrence at time t is represented by the following compact notation:

$$X_t = (S_t, R_t, D_t)$$

Here S denotes the realized state, R the relational organization through which that state is meaningful and dynamically constrained, and D the dependencies that connect the occurrence to local conditions, global organization, causal history, memory, and environment. This is an analytic decomposition, not a claim that nature contains three separately existing objects.

The formulation deliberately includes dependencies rather than treating them as optional context. A system can be locally coherent without being isolated. A cell has a membrane and regulatory organization, yet it is materially and energetically open. A social institution may have a functional boundary while depending on legal, economic, and cultural systems. A computational process may be logically individuated while depending on hardware, power, operating-system services, and stored history.

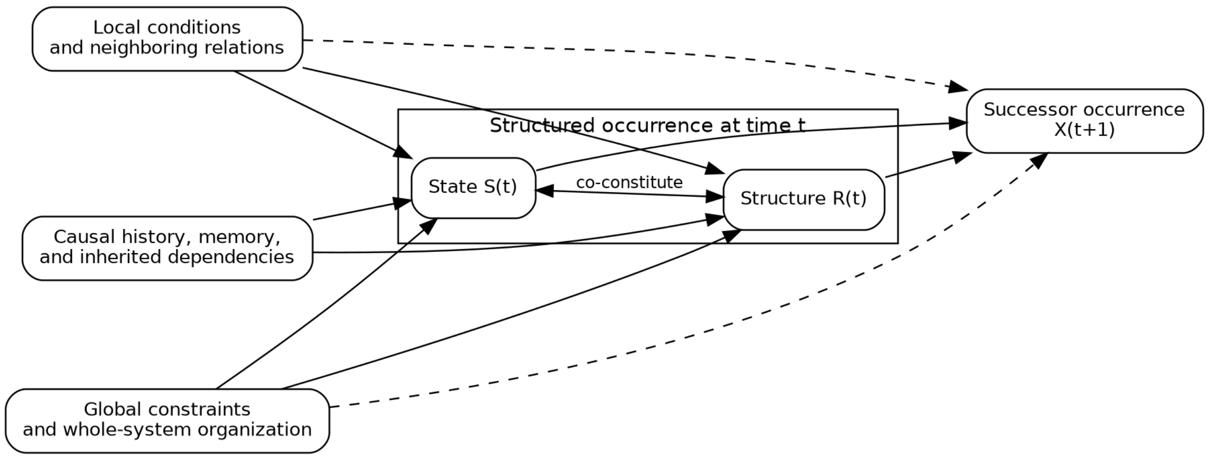

Figure 2. State and structure are analytically distinguishable but co-constituting. Local conditions, global constraints, and inherited dependencies participate in the successor transition.

No privileged decomposition is assumed. The same natural system may be modeled at molecular, cellular, organismal, ecological, computational, or institutional scales. A variable internal at one scale may be environmental at another. Different decompositions can be legitimate when they preserve different causal and explanatory regularities. The framework therefore requires every analysis to declare its scale, boundary, observation map, and intended identity question.

4. Persistence and identity

Persistence is a relation across stages, not a property of one isolated instant. The question is not whether the present stage contains a hidden token that remains numerically identical. The question is whether the present and successor stages belong to one warranted lineage under criteria appropriate to the system and inquiry.

Those criteria cannot be dictated in advance by one universal formula. Depending on the system, persistence may be tracked by material continuity, causal lineage, topology, functional organization, informational state, state-structure compatibility, transported invariants, self-maintenance, or combinations of these. A photon, a living cell, a person, a software process, and an institution need not preserve the same features. The framework is pluralist about identity criteria while insisting that they be explicit and answerable to evidence.

A natural system may instantiate objective continuity even though our criterion for tracking it is model-relative and fallible. The criterion is not thereby arbitrary. It is constrained by causal history, intervention, prediction, invariance, and robustness across measurements. What the model denies is not objective persistence, but the necessity of one absolute, context-free criterion of sameness embedded inside every thing.

A useful persistence test normally asks at least four questions. Is the later stage causally traceable to the earlier one? Does it retain organization relevant to the kind being studied? Is the realized state still compatible with its organization and conditions? Does the resulting configuration remain capable of further organized continuation? These questions form a disciplined starting point, not an exhaustive metaphysical definition.

Question	What it tests	What failure may mean
Causal traceability	Whether the successor was generated from the prior system and its dependencies.	Replacement, coincidence, or an unmodeled causal break.
Organizational continuity	Whether the organization relevant to the selected kind remains available.	Reclassification, loss of function, or a change of kind.
State-structure compatibility	Whether the realized state can be carried by the current organization.	Suppression, instability, correction, or collapse.
Viability	Whether further organized transitions remain reachable.	Terminal state, destruction, or departure from the selected identity corridor.

This account distinguishes persistence from permanence. Permanence would require a feature or bearer to remain unchanged. Persistence permits extensive change, provided a warranted continuity relation remains. It also distinguishes persistence from mere resemblance: two systems can be nearly identical without sharing a causal lineage, and one system can remain the same continuing system despite substantial internal alteration.

5. Local relations, global constraints, and wholes

The framework begins from the proposition that no element of a natural system exists or evolves in isolation. Any selected pair of elements participates in relations that are embedded in local neighborhoods and in larger organizations. Local interactions may dominate immediate transitions, while global conditions restrict the range of locally available futures. Neither level is sufficient by itself.

A whole is not a second substance placed above its parts. It is an integrated organization in which joint relations exclude some transitions, permit others, and may regenerate conditions needed for continued organization. A global constraint has explanatory force only when its path into local dynamics is explicit. Merely calculating a collective statistic does not prove that the statistic causally controls the constituents.

Wholeness must therefore be treated with the same apophatic restraint as state, structure, and relation. The word whole abbreviates a pattern of integration, constraint, and lineage. It must not become an invisible controller. Likewise, emptiness must not become a field, substrate, or final object. It is a restriction against treating any dependent organization as self-grounding.

No reified whole

The whole constrains the parts only through relations and mechanisms realized in the system. There is no additional entity named Wholeness standing outside or above those relations.

6. Constitution and description

A central unresolved issue is the distinction between what constitutes a natural system and what belongs only to a chosen description. The framework's state-structure-dependency notation cannot settle this by itself. A symbol may track a causally real organization, combine several mechanisms into one useful variable, or merely reflect a convenient coordinate choice. Formal precision does not automatically confer ontological status.

Constitution concerns the organization that makes a causal difference to the system's production, maintenance, behavior, or persistence. Description concerns the representational choices by which investigators identify variables, boundaries, states, relations, and scales. The two are connected but not identical. A good description may successfully track constitutive organization without mirroring it component by component.

Diagnostic	Evidence of a constitutive role	Evidence of a descriptive role
Intervention	Changing the factor through a controlled substrate path changes the system's trajectory or persistence.	Changing the label, coordinate, or encoding leaves the natural trajectory unchanged.
Invariance	The causal role survives reasonable changes of representation and measurement.	The feature disappears under an equivalent redescription with no empirical loss.
Prediction	Including the factor improves out-of-sample prediction under fair controls.	The factor adds notation but no discriminating power.
Mechanism tracing	A pathway connects the factor to successor dynamics or admissibility.	No physical or operational return path can be identified.
Cross-scale robustness	Related descriptions converge on the same causal dependency across scales.	The feature is confined to one arbitrary partition or coding scheme.

No single diagnostic is conclusive. Interventions can be confounded, invariances can conceal omitted structure, and multiple empirically equivalent models may remain. The proper conclusion is methodological: constitution must be argued through converging evidence, not announced by notation. A later framework may replace the present state-structure vocabulary while preserving its explanatory achievements.

This distinction also clarifies the role of presentation controls in experiments. Simultaneously relabeling both structure and state should not change the underlying system. Changing only the structure or only the state can alter their compatibility and therefore constitutes a genuine intervention. The difference between those operations is empirical and causal, not merely verbal.

7. Apophatic restraint and the ambiguity of intrinsic nature

The framework accepts the Madhyamaka criticism of independently existing self-nature as an interpretive discipline. State, structure, dependency, relation, boundary, whole, cause, model, and emptiness are all dependent distinctions. None becomes an absolute foundation merely because it performs indispensable explanatory work. This is the sense in which the framework is apophatic: every positive term remains open to challenge, replacement, and deeper contextualization.

This paper nevertheless affirms that natural systems possess real organization and causal powers. Dependence does not imply nonexistence, and model-relativity does not imply arbitrariness. The denial of independent self-nature is compatible with objective, relationally constituted persistence. A system can have a real present organization without possessing a self-grounding essence that explains itself independently of history, environment, and relation.

Much earlier ambiguity arose from the phrase intrinsic nature. In ordinary English, one may call the internal constitution of a system intrinsic even when it is changing and dependent. Jay Garfield uses the phrase in a technical Madhyamaka sense: an essence or mode of being possessed in the thing's own right, independently of anything else. The present framework does not affirm intrinsic nature in that technical sense. It therefore uses constitutive organization for the system's current, dependent organization.

Garfield's contribution is primarily a discipline against reification and a phenomenological analysis of how subjects contribute to experienced objects. Structured Becoming asks a different positive question: how does a dependently originated natural system become locally coherent, retain organized state, survive perturbation, and continue as a structured occurrence? The two projects can be complementary. The framework should learn from Garfield not that persistence is impossible, but that its own most successful categories must not be mistaken for final reality.

Apophatic rule

No explanatory object - including state, structure, relation, whole, RAIL, SHADOW, KEEPER, or emptiness - acquires independent ontological status merely because it is indispensable inside a successful model.

8. Observer dependence: a question deliberately left open

The current framework does not yet settle the place of the observer. It distinguishes a natural system from an observation produced through an interaction, but it does not decide which properties remain invariant across observers, which are interaction-relative, or how observer participation should be represented in a complete physical theory.

This restraint is especially important for Bell-type nonlocality. A globally constrained state description may offer a conceptual language for correlations that are not reducible to independent local states. That observation does not derive quantum probabilities, the Born rule, no-signaling, relativistic compatibility, or a physical mechanism of entanglement. Those questions remain separate research problems and must not be decided by metaphysical preference.

The framework should eventually include observers as natural systems with their own states, structures, histories, instruments, and coupling relations. Doing so must avoid two extremes: treating observations as passive windows onto prepackaged properties, and collapsing natural systems into products of consciousness. For now, this is an explicit open boundary rather than a hidden assumption.

9. A compact formal sketch

The original working paper placed formal definitions at the center. The present revision reverses that priority. Formalism is retained as a discipline and research instrument, but it serves the philosophical account rather than replacing it.

At a declared scale and boundary, the model represents the current occurrence through state, relational organization, and dependencies. A general transition is written:

$$X_{t+1} = F(X_t, L_t, G_t, E_t, \Delta_t)$$

The additional terms denote local conditions, global constraints, environmental input, and a represented or implemented change. These distinctions are model-relative. A descriptive difference between two states is not automatically a causal force; it becomes causal only when a declared mechanism uses it in generating the successor.

Persistence is represented by a kind-relative relation rather than one metaphysically universal predicate:

$$P_K(X_t, X_{t+1} \mid \sigma, \tau)$$

The subscript identifies the kind or identity question; the conditioning terms identify the selected scale, observation scheme, and declared tolerances. A practical implementation may aggregate causal traceability, organizational continuity, compatibility, viability, and domain-specific invariants. The formal appendix gives one conjunctive implementation, but the framework does not insist that every domain use it unchanged.

Three pieces of project vocabulary remain useful when treated cautiously. A RAIL is a declared corridor of viable continuation. A SHADOW is a measured component of proposed change that misses that corridor, not an occult realm. A KEEPER is an explicit policy that selects pass-through, correction, or environmental refresh. None is a universal law or inner agent; each names a possible mechanism within a particular model.

Term	Model role	Interpretive restriction
RAIL	A viable region, trajectory, manifold, attractor neighborhood, or other corridor of admissible continuation.	Not a pre-existing cosmic track or final geometry.
SHADOW	Residual, prediction error, incompatibility, or inadmissible component under a declared metric.	Not a hidden substance, force, or second realm.
KEEPER	An auditable selection policy among available update paths.	Not a homunculus or universal controller.
Delta	A sufficient represented difference or an implemented update input.	Description and causal input must not be silently conflated.

10. Structure-state compatibility as a testable claim

The most concrete formal result concerns the compatibility between a relational operator and a realized state. A structure may contain long-lived modes that the current state barely excites. Conversely, a state with substantial magnitude may decay rapidly when its organization channels it into transient modes. Persistence therefore depends on their alignment, not on either inventory considered alone.

For a normal linear operator and an initial state expanded in its eigenbasis, the exact identity is:

$$\|A^t x_0\|^2 = \sum_i |\lambda_i|^{2t} \, |v_i^H x_0|^2$$

The result is mathematically modest but philosophically important. Persistent possibilities in the structure matter only when the realized state overlaps them. Simultaneously relabeling structure and state preserves the relationship; changing only one generally changes the trajectory. The theorem does not establish the nonlinear, stochastic, or topology-changing cases, but it isolates a mechanism that can be tested under controlled interventions.

This formal example should not be promoted into a universal theory of persistence. Its value is methodological: it demonstrates how a broad metaphysical claim can be translated into a precise, falsifiable relation while keeping the assumptions visible.

11. Research program and conditions of revision

Structured Becoming becomes scientifically useful only when it produces discriminating tests. The framework should be evaluated against simpler state-only, structure-only, and history-only alternatives. It should also be challenged across alternative scales and decompositions rather than protected by one favored representation.

1. Register the natural system, selected scale, boundary, observation map, and identity question before testing persistence.
2. Vary state while holding structure and relevant conditions fixed; then vary structure while holding state fixed.
3. Use simultaneous relabeling as a presentation control and one-sided changes as compatibility interventions.
4. Trace every claimed local or global causal influence through an explicit substrate or transition path.
5. Compare multiple decompositions and report whether the persistence claim is robust, transformed, or lost across them.
6. Separate descriptive deltas, feedback corrections, projections, penalties, and environmental refresh mechanisms.
7. Retain failed, singular, and null cases and apply adversarial tripwires against post hoc identity criteria.

The framework should be revised or abandoned in a declared domain if state-structure compatibility adds no discriminating power, if its causal categories cannot be tied to distinct mechanisms, if predictions depend entirely on an arbitrary encoding, or if simpler models perform equivalently under fair controls.

12. Open problems

The theory remains intentionally incomplete. Its open problems are not concealed gaps but the agenda by which it must improve.

Problem	Question
Individuation	What evidence justifies treating a selected process as one system rather than a convenient aggregate?
Scale and decomposition	Which decompositions are causally informative, and when can several non-equivalent decompositions be equally legitimate?
Identity thresholds	When does transformation become replacement or a change of kind?
Constitution and description	Which model variables track constitutive organization, and which are presentation-dependent conveniences?
Observer dependence	Which properties and boundaries remain stable across observers, instruments, and encodings?
Bell-type nonlocality	Can global relational constraints illuminate quantum correlations without violating no-signaling or relativistic requirements?
Novelty	Can genuinely new organizational possibilities arise, or are all futures implicit in a prior state space?
Temporal direction	What grounds the asymmetry between retained history and unrealized future?
Normativity	When do viability, error, and correction belong to the organization rather than to an external evaluator?
Physical realization	Which formal terms correspond to physical mechanisms in a given domain?
Apophatic productivity	How can the research program accumulate knowledge without reifying its best current formalism?

13. Relationship to neighboring traditions

Structured Becoming is a candidate synthesis. It does not claim to have invented process metaphysics, persistence theory, relational ontology, cybernetics, autopoiesis, structural realism, or Madhyamaka.

It agrees with process philosophy that becoming and organized continuity are more fundamental to persistence than an unchanging substance, while refusing to impose one universal process entity across every domain. It agrees with stage and perdurance theories that diachronic identity is constructed across temporally distinct stages, while remaining pluralist about the criteria that warrant that construction.

It agrees with relational and structural approaches that organization is indispensable to what entities are and how they behave. It rejects the stronger inference that structure itself is the one independently existing reality. It draws on cybernetics and autopoiesis for explicit mechanisms of feedback, closure, and self-maintenance, while recognizing that many systems are co-produced across open boundaries.

Madhyamaka contributes the anti-reification discipline. Garfield's presentation of dependent origination and emptiness helps prevent the positive model from becoming an ultimate ontology. Structured Becoming contributes a complementary positive project: a revisable account of how dependent natural systems achieve organized continuity.

14. Core commitments

- Natural systems are real, but no model is identical with the natural system it represents.
- No selected entity exists in complete isolation from local, global, historical, and environmental relations.
- At a declared scale, persistence is modeled through the carryover of state, relational structure, and dependencies.
- State and structure are analytically distinguishable but mutually dependent and jointly realized.
- Persistence does not require permanence, an immutable bearer, or an independently existing essence.
- Natural systems may instantiate objective continuities even though model criteria are fallible, scale-relative, and revisable.
- No single identity criterion or decomposition is assumed to be universally privileged.
- Constitution cannot be read directly from notation; it must be supported by intervention, invariance, prediction, and mechanism tracing.
- Wholes, relations, and emptiness must not be reified into hidden substrates or controllers.
- Observer dependence and Bell-type nonlocality remain open questions rather than conclusions of the framework.
- Formalism is retained for discipline and testability, but it does not replace philosophical explanation.
- Every category in the framework, including the framework itself, remains open to revision or abandonment.

Conclusion

The question "What persists?" does not require an immutable essence. It requires an account of how a present organization becomes causally available to a successor. Structured Becoming proposes that a natural system continues through a relationally and historically produced organization of state, structure, and dependencies. The organization is real and causally consequential, but it is neither isolated nor self-grounding.

The framework is deliberately asymmetric about reality and representation. Natural systems are not created by our models, yet every model is abstract, selective, and revisable. State, structure, dependency, boundary, and identity are tools for tracking objective organization; they are not guaranteed to be reality's final vocabulary. Multiple decompositions may disclose different real regularities, and the distinction between constitution and description must be earned empirically and philosophically.

The resulting position is neither substance metaphysics nor nihilism. It is a positive account of dependent persistence constrained by an apophatic refusal to absolutize its own concepts: a thing persists as organized continuity through transformation, while every term used to describe that continuity remains dependent, revisable, and open to deeper explanation.

Appendix A. Minimal formal apparatus

A.1 Structure-dependent state space

Let R range over relational organizations. The states meaningful under a given R form its associated state space. The total modeled configuration space is:

$$\mathcal{Q} = \{(R, S, D) : R \in \mathcal{R},\ S \in X_R,\ D \in \mathcal{D}\}$$

This typing represents co-constitution. It does not prove that natural reality is literally fibered or that the decomposition is unique.

A.2 One operational persistence predicate

For a selected kind K and declared tolerances, one strict implementation is:

$$P_t = H_t \wedge O_t \wedge C_t \wedge V_t$$

The four terms test historical traceability, organizational continuity, state-structure compatibility, and successor viability. The conjunction is a demanding operational choice, not a universal metaphysical law. Other domains may use graded aggregation, partial ordering, or different criteria.

A.3 RAIL, SHADOW, and KEEPER

Let the admissible set collect configurations allowed under the current conditions, and let the error function be a declared incompatibility measure. A RAIL can be represented as the viable corridor:

$$\mathrm{RAIL}_\tau(C_t) = \{X \in \mathcal{A}(C_t) : \varepsilon(X, C_t) \leq \tau\}$$

For a smooth local model, a proposed update may be decomposed into an admissible component and a residual component. The latter is the SHADOW under that metric. Broader uses of SHADOW must specify whether they mean constraint violation, prediction error, numerical residual, or unmodeled structure.

One auditable KEEPER policy uses flags L = low residual, H = high residual, R = regular constraints, and S = stale environment:

$$g_{\mathrm{refresh}} = H \vee S \vee \neg R$$

$$g_{\mathrm{pass}} = L \wedge \neg S \wedge R$$

$$g_{\mathrm{correct}} = \neg L \wedge \neg H \wedge R \wedge \neg S$$

Condition	**Selected action**
High residual, stale environment, or irregular constraints	Environmental refresh or declared external re-alignment
Low residual, fresh environment, regular constraints	Pass the proposed update
Intermediate residual, fresh environment, regular constraints	Apply an explicit internal correction

The circuit is included for auditability. It is an engineering mechanism that may instantiate the broader framework; it is not part of the definition of every persisting thing.

Appendix B. Structure-state alignment result

Let A be a normal linear operator and expand the initial state in its orthonormal eigenbasis. Then:

$$\|A^t x_0\|^2 = \sum_i |\lambda_i|^{2t} |c_i|^2$$

Proof. Under repeated application of A, each eigenvector component is multiplied by the corresponding eigenvalue raised to the t-th power. Orthogonality removes cross terms when the squared norm is taken. The identity follows.

The result shows that long-lived modes in the operator do not guarantee a long-lived realized trajectory. The initial state must overlap those modes. A simultaneous permutation of the operator and initial state preserves that overlap, while changing only one generally changes it. This separates presentation changes from genuine compatibility interventions.

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